INT 303 BIG DATA ANALYTICS

#Lecture12: Summary & Review Part 2

Jia Wang
Jia.wang02@xjtlu.edu.cn

Xi'an Jiaotong-Liverpool University
西交利物浦大學

FINAL EXAMINATION

Reading Time: *5 mins*

Writing Time: *120 mins*

INSTRUCTIONS TO CANDIDATES

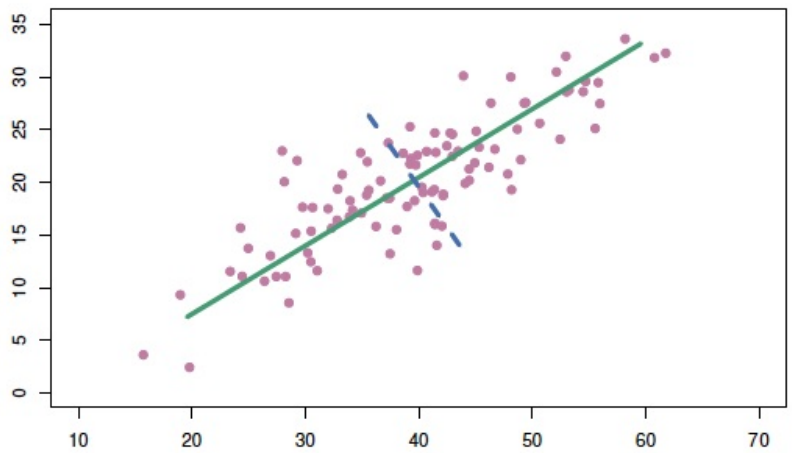
1. **This is a closed-book examination. NO notes or books are permitted.**
2. **Total marks available are 100. The number on the right indicates the mark for each question.**
3. **Attempt ALL questions. Write all the answers in the answer booklet provided.**
4. **Only solutions written in English will be accepted.**
5. **Correct answers do not guarantee a full score: mark penalties may be imposed for missing intermediate solution steps or illogical solution processes.**

Questions:

5. Principal Components Analysis

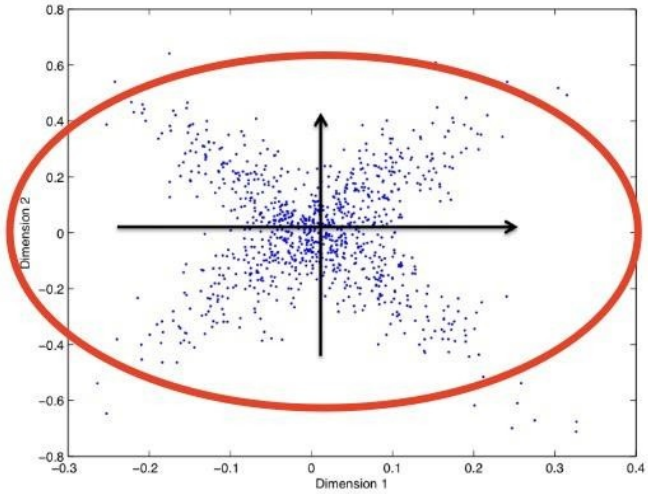
The Intuition Behind PCA

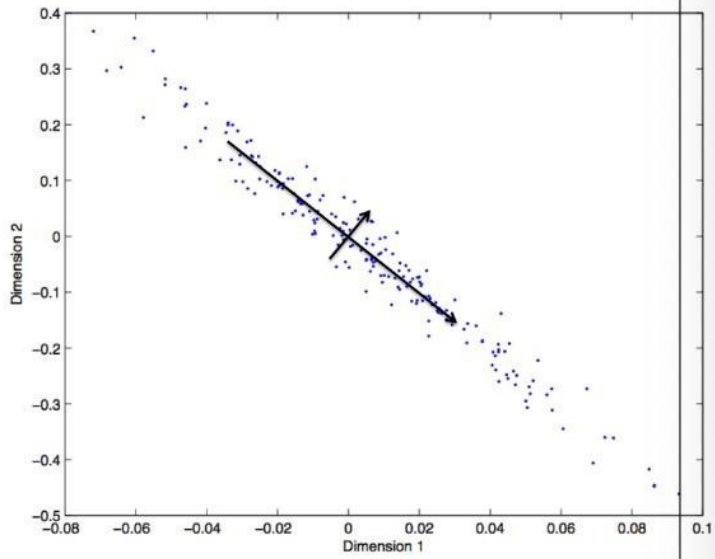
- Top PCA components capture the most of amount of variation (interesting features) of the data.
- Each component is a linear combination of the original predictors - we visualize them as vectors in the feature space.



EXERCISES

Solution:





Questions:

6. Large-scale Machine Learning Linear Regression
Logistic Regression
Decision Tree
Nearest Neighbor Classifier
Support Vector Machines
Convolutional Neural Network K-means
...

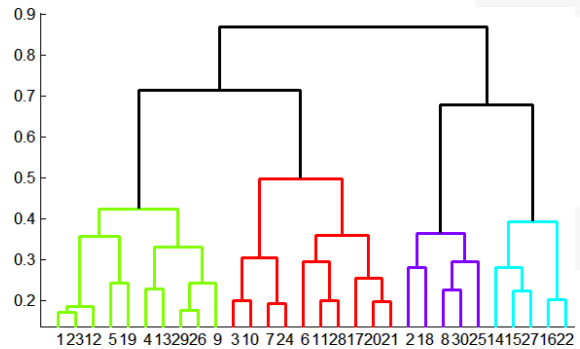
You should identify one model or technique that best addresses the practical problem and provide a justification for why it is the most suitable choice.

Questions:

7. Clustering

...

HIERARCHICAL CLUSTERING



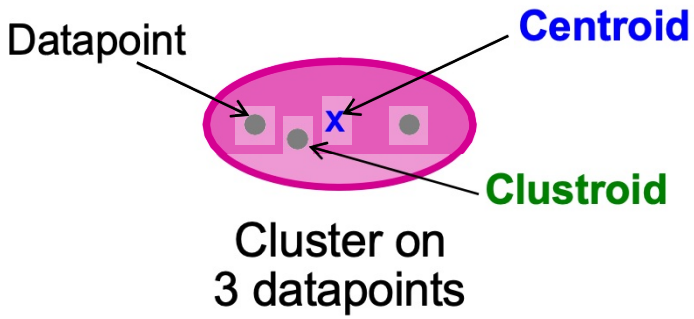
- **Key operation:**
  Repeatedly combine two nearest clusters

- **Three important questions:**
  - **1)** How do you represent a cluster of more than one point?
  - **2)** How do you determine the “nearness” of clusters?
  - **3)** When to stop combining clusters?

EUCLIDEAN SPACE

- **Key operation: Repeatedly combine two nearest clusters**
- **(1) How to represent a cluster of many points?**
  - **Key problem:** As you merge clusters, how do you represent the "location" of each cluster, to tell which pair of clusters is closest?
- **Euclidean case:** each cluster has a ***centroid*** = average of its (data)points
- **(2) How to determine "nearness" of clusters?**
  - Measure cluster distances by distances of centroids

CENTROID AND CLUSTROID

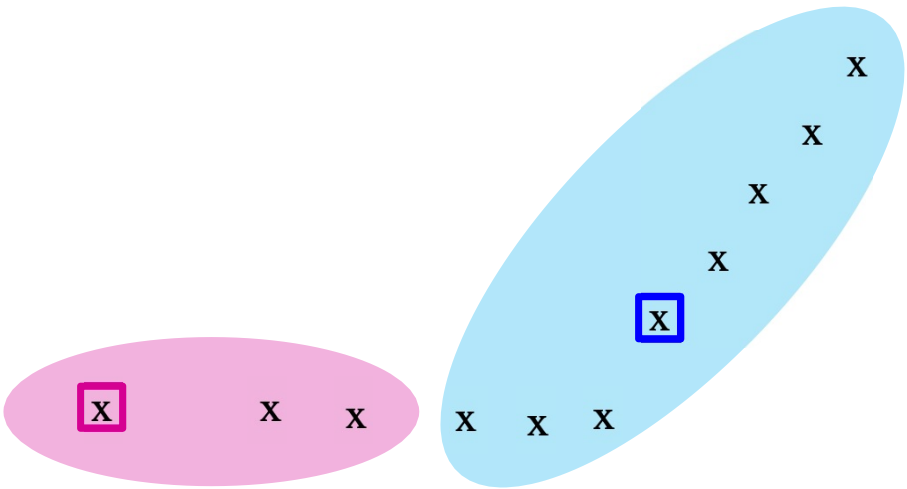


Centroid is the avg. of all (data)points in the cluster. This means centroid is an "**artificial**" point.

Clustroid is an **existing** (data)point that is "closest" to all other points in the cluster.

Example: Assigning Clusters

***k* = 2**

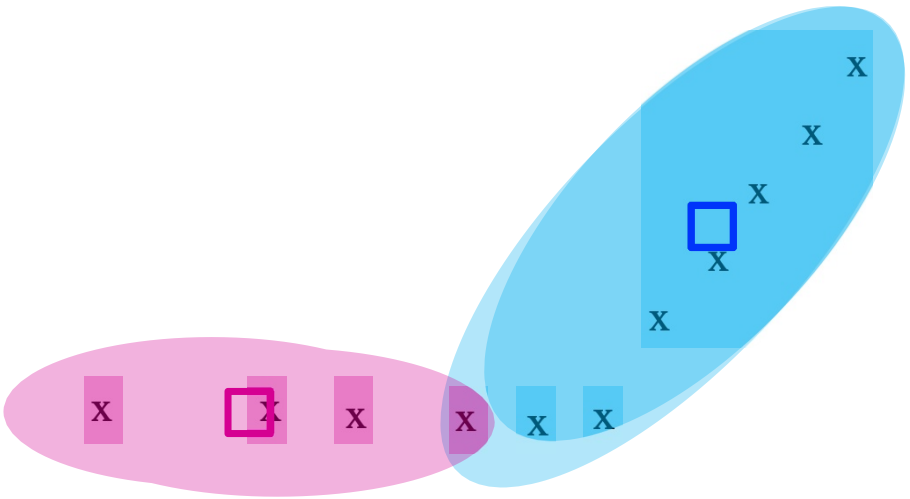


x … data point
□ … centroid

Clusters after round 1

Example: Assigning Clusters

***k* = 2**

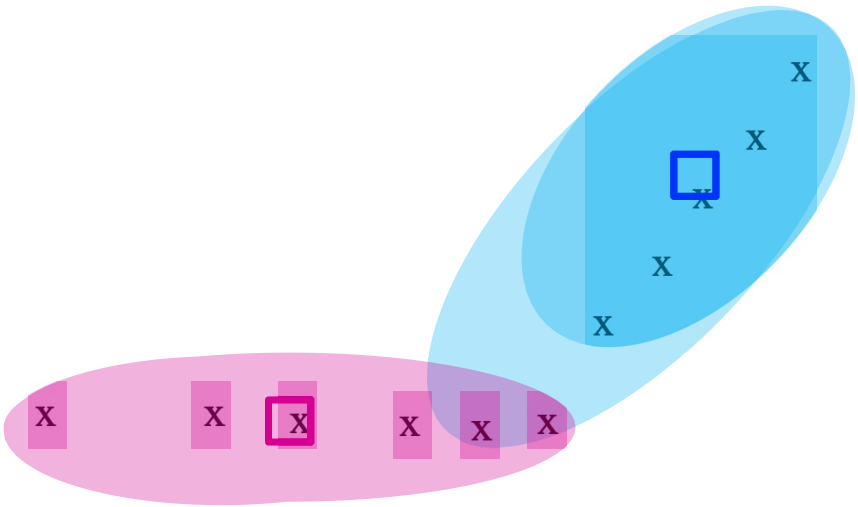


x … data point
□ … centroid

Clusters after round 2

Example: Assigning Clusters

***k* = 2**

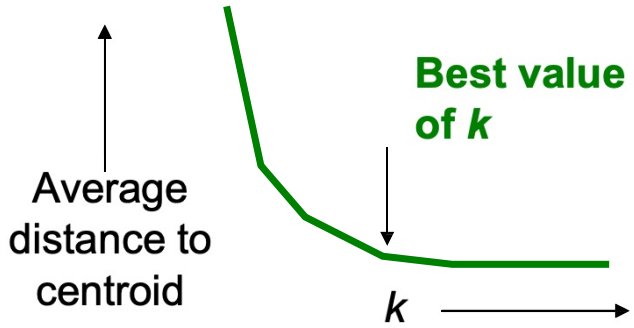


x … data point
□ … centroid

Clusters at the end

GETTING THE *K* RIGHT

- Average falls rapidly until right **k**, then changes little



Questions:

8. Recommendation Systems Content-based Systems Collaborative Filtering

User-user CF

Item-item CF

…

FORMAL MODEL

- X = set of **Customers**
- S = set of **Items**

- **Utility function** $u: X \times S \rightarrow R$
  - R = set of ratings
  - R is a totally ordered set
  - e.g., **0-5** stars, real number in **[0,1]**

KEY PROBLEMS

- **(1) Gathering "known" ratings for matrix**
  - How to collect the data in the utility matrix

- **(2) Extrapolate unknown ratings from the known ones**
  - Mainly interested in high unknown ratings
    - We are not interested in knowing what you don't like
      but what you like
- **(3) Evaluating extrapolation methods**
  - How to measure success/performance of recommendation methods

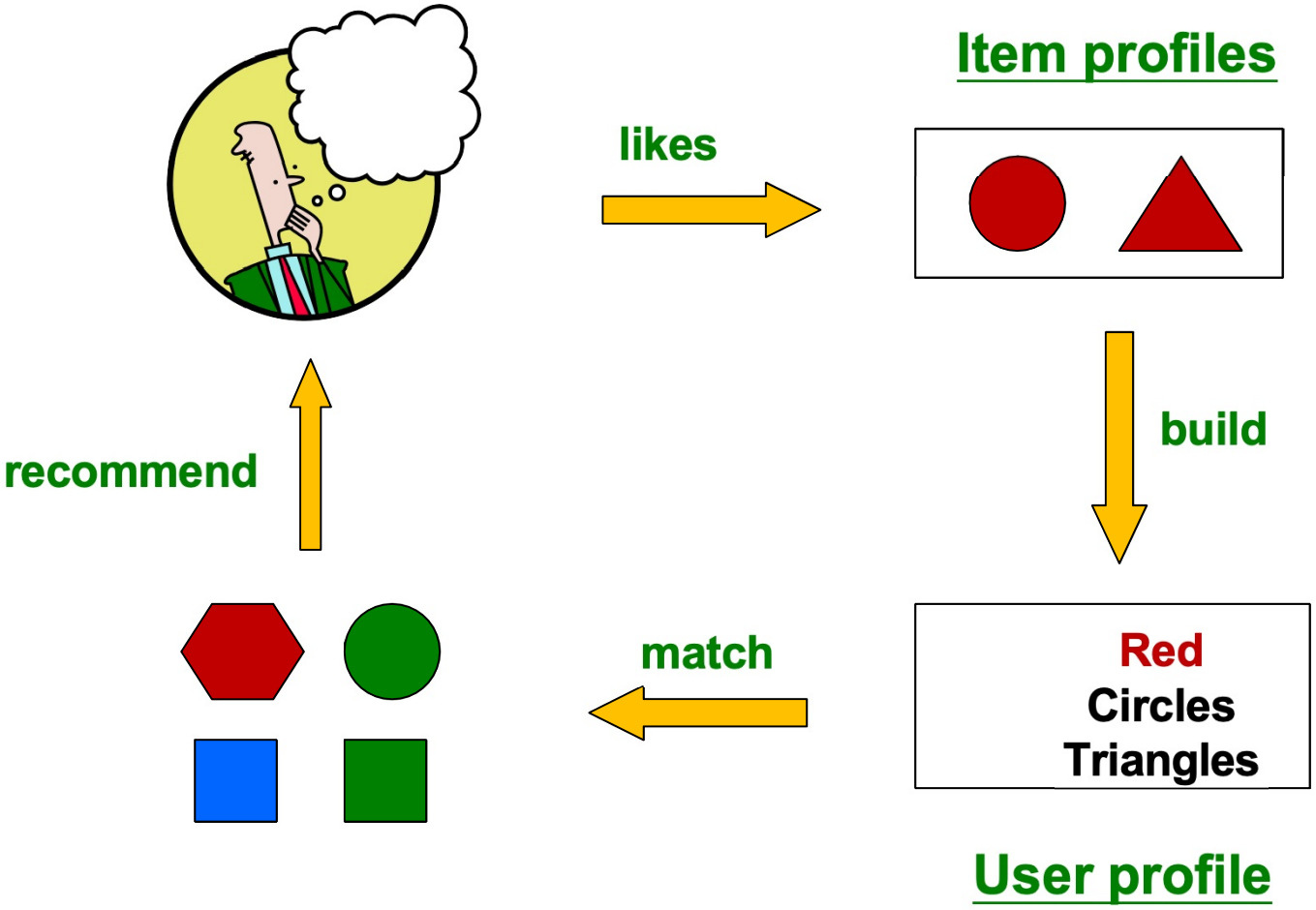
CONTENT-BASED RECOMMENDATIONS

- **Main idea:** Recommend items to customer ***x*** similar to previous items rated highly by ***x***

Example:

- **Movie recommendations**
  - Recommend movies with same actor(s), director, genre, …
- **Websites, blogs, news**
  - Recommend other sites with “similar” content

PLAN OF ACTION



ITEM PROFILES

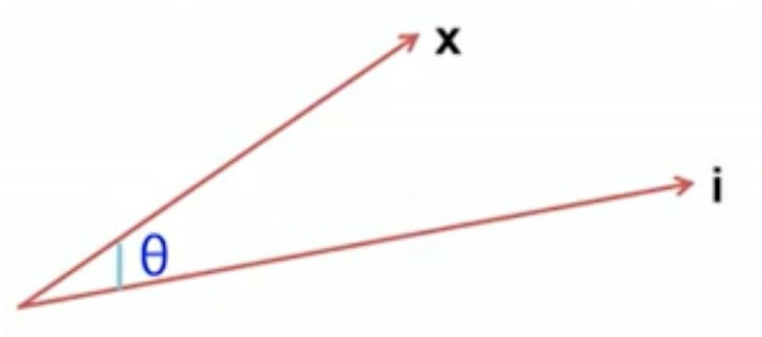
- For each item, create an **item profile**
- **Profile is a set (vector) of features**
  - **Movies:** author, title, actor, director,…
  - **Text:** Set of “important” words in document

USER PROFILES

- **User profile possibilities:**
  - Weighted average of rated item profiles
  - **Variation:** Normalize weights using average rating of user
  - More sophisticated aggregations possible…

COSINE SIMILARITY

- Estimate $u(\boldsymbol{x}, \boldsymbol{i}) = \cos(\boldsymbol{x}, \boldsymbol{i}) = \frac{\boldsymbol{x} \cdot \boldsymbol{i}}{||\boldsymbol{x}|| \cdot ||\boldsymbol{i}||}$



For convenience, we use $\cos(\theta)$ as our similarity measure and call it the "cosine similarity" in this context.

PROS: CONTENT-BASED APPROACH

- **+: No need for data on other users**
  - No (item) cold-start or sparsity problems
- **+: Able to recommend to users with unique tastes**
- **+: Able to recommend new & unpopular items**
  - No first-rater problem
- **+: Able to provide explanations**
  - Can provide explanations of recommended items by listing content-features that caused an item to be recommended

CONS: CONTENT-BASED APPROACH

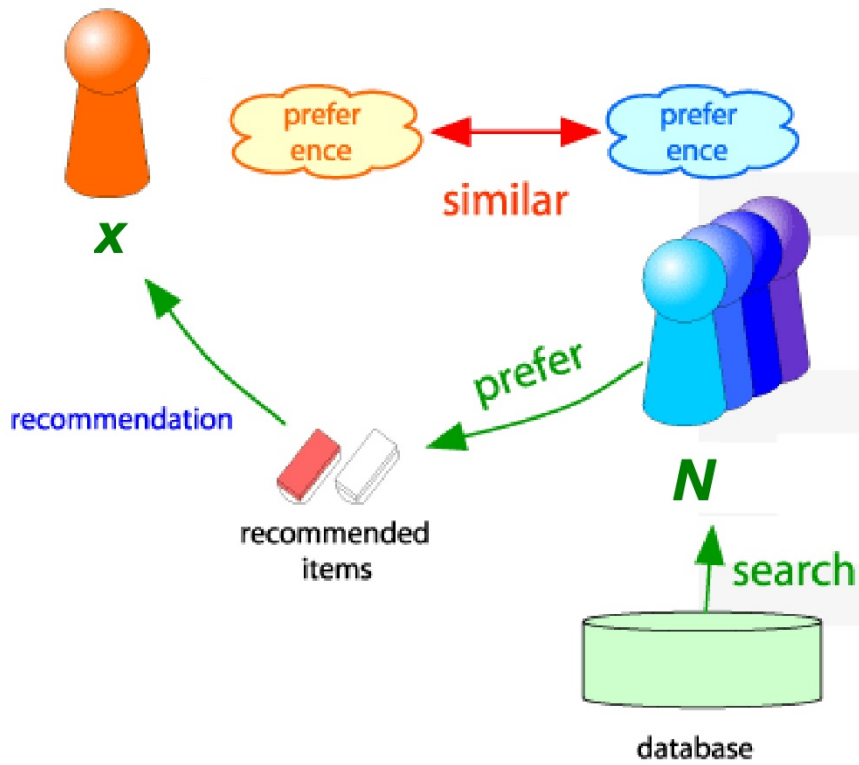
- **–: Finding the appropriate features is hard**
  - E.g., images, movies, music
- **–: Overspecialization**
  - Never recommends items outside user's content profile
  - People might have multiple interests
  - **Unable to exploit quality judgments of other users**
- **–: Recommendations for new users**
  - **How to build a user profile? (User) cold-start problem**

COLLABORATIVE FILTERING

- Consider user ***x***

- Find set ***N*** of other users whose ratings are "**similar**" to ***x***'s ratings

- Estimate ***x***'s ratings based on ratings of users in ***N***



ITEM-ITEM CF (|N|=2)

users

movies

	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3			5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

- unknown rating

- rating between 1 to 5

ITEM-ITEM CF (|N|=2)

users

movies

	1	2	3	4	**5**	6	7	8	9	10	11	12
1	1		3		?	5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

- estimate rating of movie **1** by user **5**

ITEM-ITEM CF (|N|=2)

users

movies

	1	2	3	4	**5**	6	7	8	9	10	11	12	sim(1,m)
1	1		3		?	5			5		4		**1.00**
2			5	4			4			2	1	3	**-0.18**
3	2	4		1	2		3		4	3	5		**0.41**
4		2	4		5			4			2		**-0.10**
5			4	3	4	2					2	5	**-0.31**
6	1		3		3			2			4		**0.59**

Neighbor selection:
Identify movies similar to movie **1**, **rated by user 5**

Here we use mean-adjusted cosine similarity:

1) Subtract mean rating m_i from each movie i

$m_1 = (1+3+5+5+4)/5 = 3.6$

row 1: *[-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4, 0, 0.4, 0]*

2) Compute cosine similarities between rows

3) Predict by taking weighted average

ITEM-ITEM CF (|N|=2)

users

movies

	1	2	3	4	5	6	7	8	9	10	11	12	sim(1,m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		0.41
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		0.59

Compute similarity weights:

$s_{1,3}=0.41$, $s_{1,6}=0.59$

ITEM-ITEM CF (|N|=2)

users

movies

	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3		2.6	5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

Predict by taking weighted average:

$$r_{ix} = \frac{\sigma_{j \in N(i;x)} s_{ij} \cdot r_{jx}}{\sigma s_{ij}}$$

$r_{1.5}$ = (0.41*2 + 0.59*3) / (0.41+0.59) = 2.6